

Somaia Vidyavihar University
K J Somaia School of Engineering
Department of Electronics and Telecommunication Engineering

Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 1
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Experiment No: 1

Title: Database application description and Design of Extended-Entity-Relationship diagram

Objective: To define a Database Problem and Design an ER diagram for a business domain.

Expected Outcome of Experiment:

CO1. Understand the role of DBMS in real-world applications and differentiate it from traditional file systems.

CO2. Understand various data models and apply ER modeling to design diagrams for real-world scenarios.

Books/ Journals/ Websites referred:

1. G. K. Gupta : "Database Management Systems", McGraw – Hill
 2. Korth, Silberchatz, Sudarshan : "Database Systems Concept", 6th Edition , McGraw Hill
 3. Elmasri and Navathe, "Fundamentals of Database Systems", 5th Edition, PEARSON Education.
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Dia Software: A software to Design ER Model

Dia is one of the convenient open source tool which runs on multiple platforms including Linux, Windows and MacOS. Dia has a number of "sheets" each of which includes diagram objects for different modeling tools, such as UML, ER diagrams, flowcharts, etc.

The ER tool has objects for entities, relationships, attributes (using the oval notation), edges, and so on. The properties boxes for each of these elements allow you to specify cardinality

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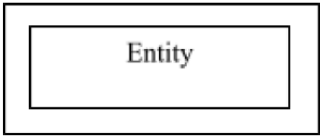
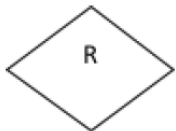
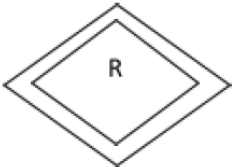
constraints, total participation, identifying relationship, etc. It supports many common formats to store diagrams such as jpeg, png, eps, etc.

Pre Lab/ Prior Concepts:

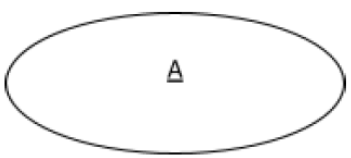
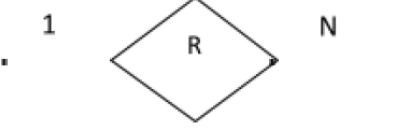
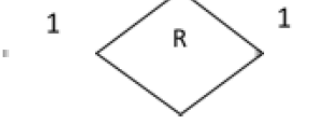
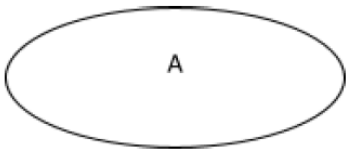
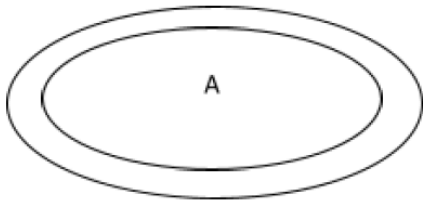
ER Model:

The ER data model was developed to facilitate the database design by allowing specification of an enterprise schema that represents the overall logical structure of the database. The ER model is one of the several data models. The semantic aspect of the model lies in its representation of the meaning of the data. The ER model is very useful many database design tools drawn on concepts from the ER model. The ER model employs 3 basic notations: entity set, relationship set and attributes.

Symbols Used in ER Notation

Sr. No.	Diagram	Explanation
1.		Entity set: An entity is a set of entities of the same type that share the properties or attributes.
2.		Weak entity set: An entity set may not have sufficient attributes to form a primary key. Such an entity set is termed as weak entity set
3.		Relationship Set: A relationship is an association among several entities. A relationship set is a set of relationship of the same type.
4.		Identifying relationship set for weak entity set: The relationship associating the weak entity set with the

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5.		Primary key: The primary key is used to denote a candidate key that is chosen by the database designers as the principal means of identifying entities within an entity set.
6.		Many to Many Relationship
7.		One to One relationship
8.		Attribute
9.		Multi valued

Extended Entity Relationship Diagram:

The EER model includes all of the concepts introduced by the ER model. Additionally it includes the concepts of a subclass and superclass (Is-a), along with the concepts of specialization and generalization. Furthermore, it introduces the concept of a union type

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or category, which is used to represent a collection of objects that is the union of objects of different entity types. EER model also includes EER diagrams that are conceptual models that accurately represent the requirements of complex databases.

Sample Database application Description:

Example Case Study: List the data requirements for the database of the company which keeps track of the company employee, department and projects. The database designers provide the following description

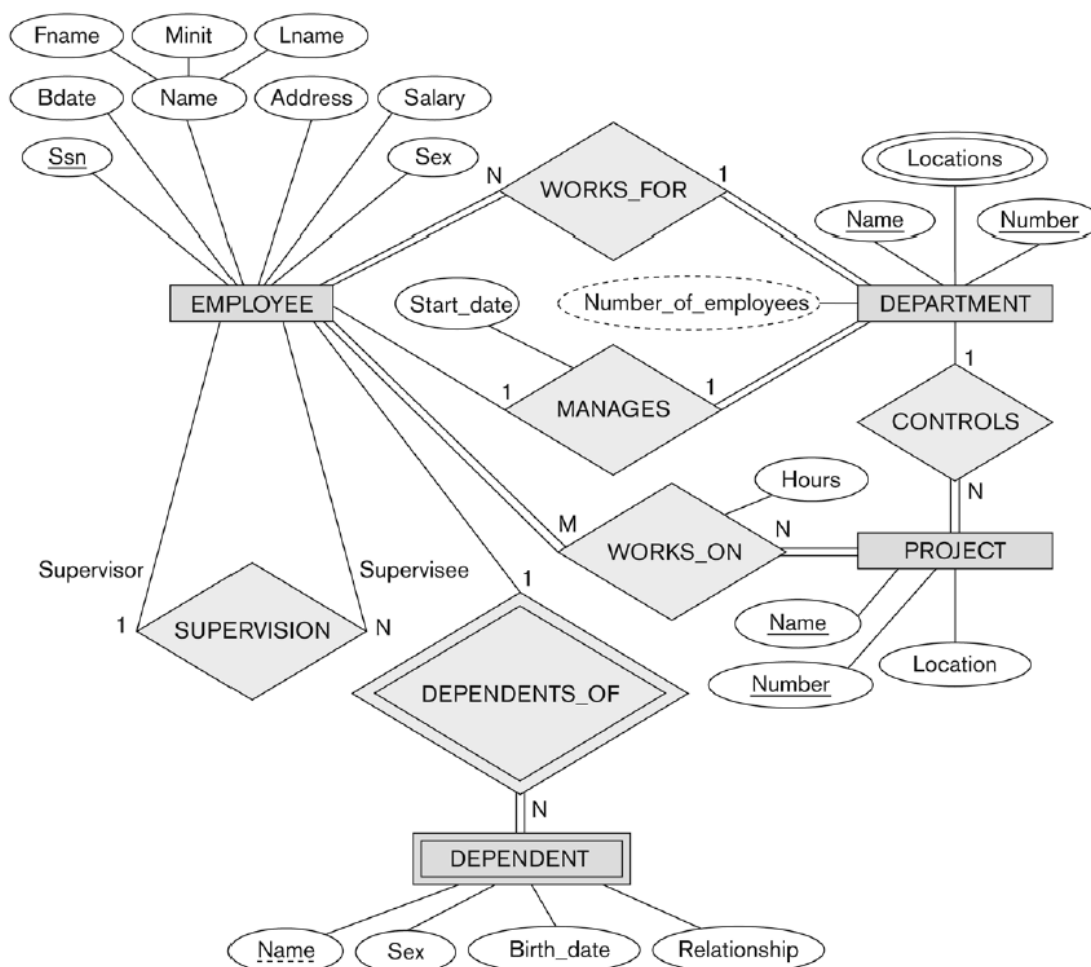
1. The company is organized into departments. Each department has unique name, unique number, and particular employee to manage the department. We keep track of the start date and the employee begins managing the department. The department has several locations.
2. The department controls a number of projects each of which has a unique name, unique number and a single location.
3. We store each employee names social security number, address, salary, sex and dob. An employee is assigned one department but may work on several projects which are not necessarily controlled by the same department. We keep track of the department of each employee works on each project and for insurance purpose. We keep each dependents first name, gender, date of birth and relation

Procedure for doing the ER diagram experiment

1. Identifying the Entities (Strong and weak entities)
2. Identify attributes of the Entity (keys, partial key, simple, composite, multivalued, derived)
3. Identify relationship(recursive)
4. Identify the structural constraints of the relationship (cardinality ratio, participation constraints)

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Sample ER- Diagram for company Case Study Database:



E R Diagram of Company Database

Database application description :

Title of the Database application:

#Include the following points

Aim of the Database Application (Purpose of the Database application)

Users of the Application (Target audience who will interact with your application)

Data Stored in the Database(Identify the key entities and their attributes and relationship)

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Analysis Capabilities of the Database (Specify the potential analyses and insights that users can derive from the data stored in the database)

Entity names with attributes:

1. Customer

Customer_ID (Primary Key)

Name

Address

2. Account

Account_Number (Primary Key)

Account_Type

Open_Date

Balance

3. Branch

Branch_ID (Primary Key)

Branch_Name

City

IFSC_Code

4. Employee

Employee_ID (Primary Key)

Employee_Name

Designation

5. Transaction

Transaction_ID (Primary Key)

Transaction_Type

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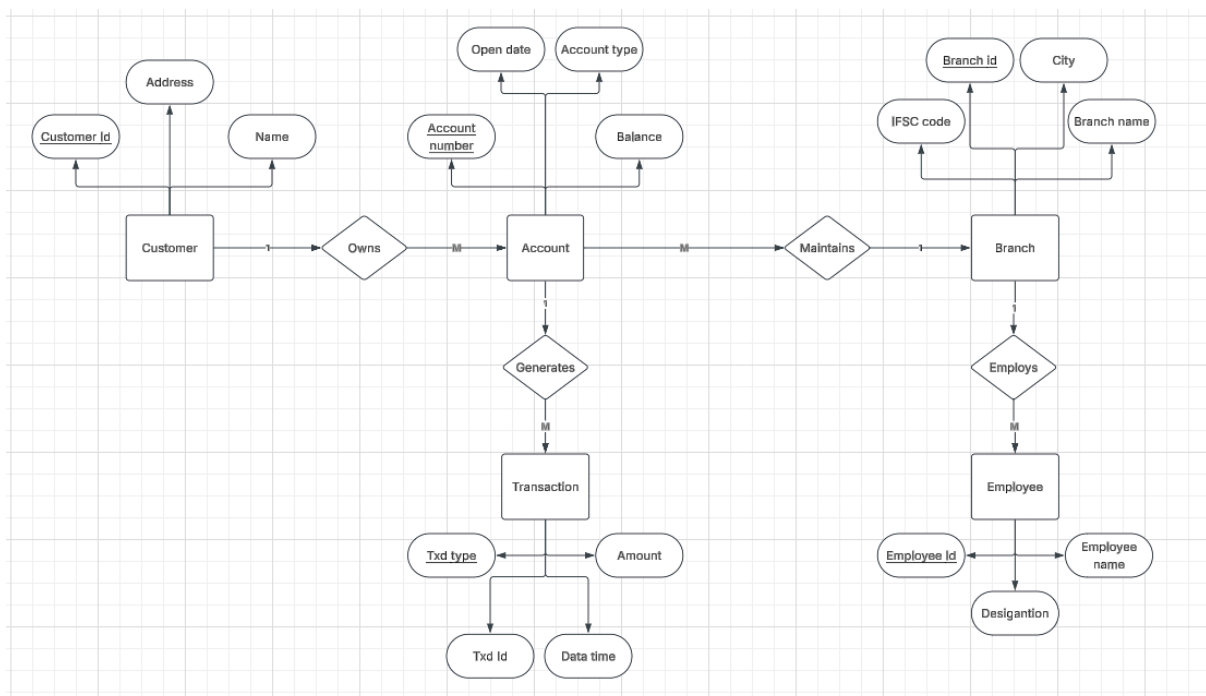
Amount

Date_Time

Relationships names related entities :

Relationship Name	Related Entities	Cardinality
Owns	Customer – Account	1 : M
Maintains	Branch – Account	1 : M
Generates	Account – Transaction	1 : M
Employs	Branch – Employee	1 : M

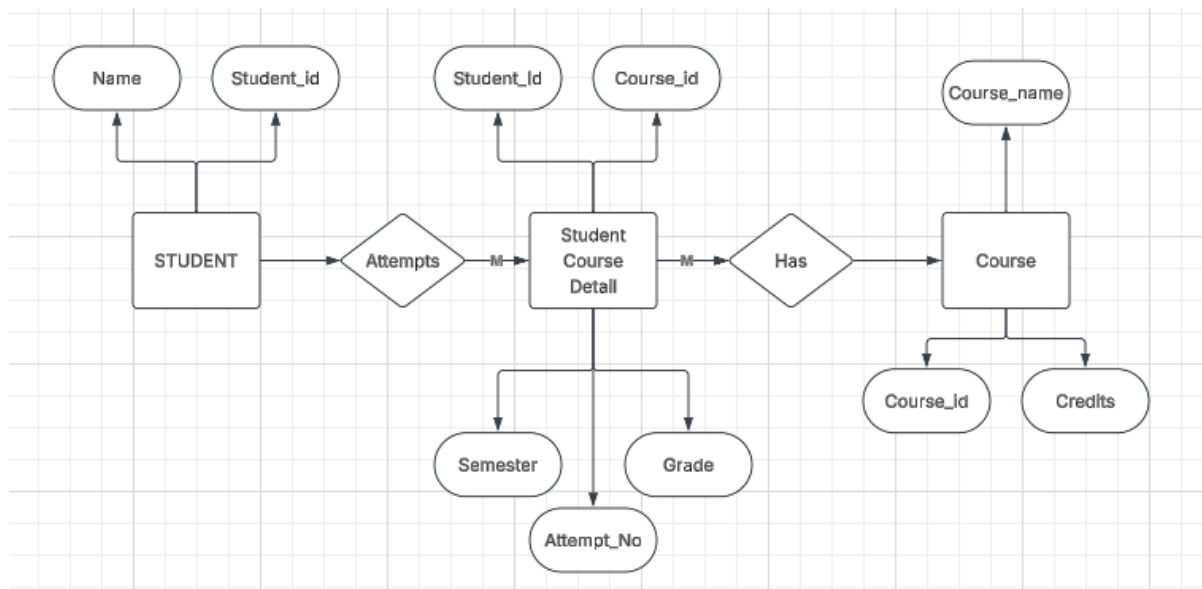
EER diagram of chosen Database application :



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Post Lab Descriptive Questions: `

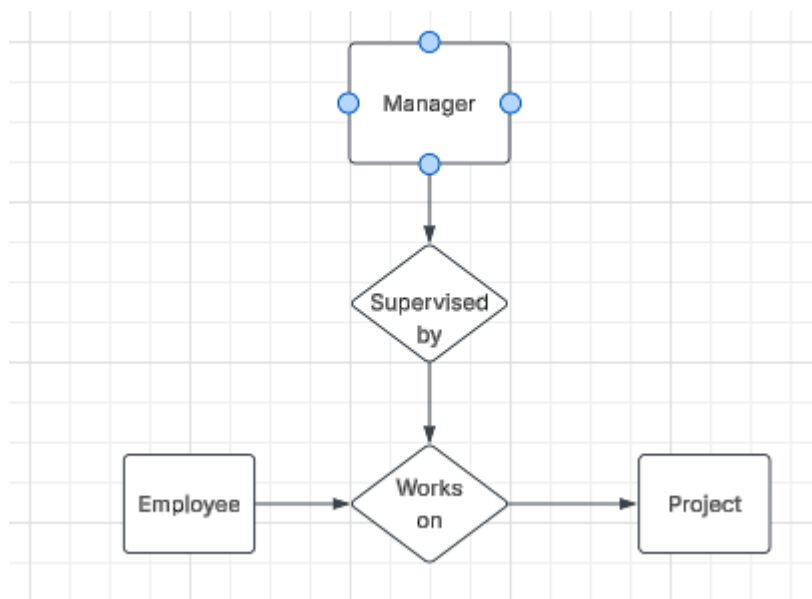
1. In the Academic database a Grade is issued to each STUDENT for each COURSE taken and stored in the STUDENT COURSE DETAIL entity. A STUDENT may decide to re-take a COURSE to better their GRADE. The administration would like to keep a record of the old/previous Grade as well as the new Grade. Show ER diagram to include historical Grades if the students should have them.



To preserve both previous and new grades, create a separate entity called STUDENT_COURSE_DETAIL. Each attempt at a course is stored as a separate record with an Attempt_No. This allows the database to maintain the complete grade history without overwriting previous grades.

2. Discuss the concept of aggregation. Give an example. How to represent aggregation in ER model
Aggregation is an abstraction mechanism in the ER model that allows a relationship set to be treated as a higher-level entity set.
It is mainly used when a relationship itself participates in another relationship.
It is needed because Standard ER models do not allow relationships to have relationships.
Aggregation overcomes this limitation by grouping an entity-relationship pair into a single abstract entity.

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Aggregation is an abstraction mechanism in the Entity-Relationship (ER) model in which a relationship set is treated as a higher-level entity so that it can participate in another relationship. It is useful when a relationship itself needs to be associated with another entity.

Aggregation is required because a standard ER model allows relationships only between entities, not between another relationship and an entity.

3. Two separate banks which decide to merge. Both banks use same ER database schema (Assume the ER diagram). If the merged bank is to have a single database, there are several potential problems:

- The possibility that two original banks have branches with the same name
- The possibility that some customers are customers of both original banks
- The possibility that some loan or account numbers were used at both original banks

Discuss for each of these potential problems, why there is indeed potential difficulty in database based on ER model. Propose a solution to a problem. For your solution, explain any changes that would have to be made and describe what their effect would be on the ER database schema and the data.

Assign every branch a unique Branch_ID and use it as the primary key. Branch names can remain the same because the Branch_ID uniquely identifies each branch.

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Generate new unique account and loan numbers or prefix existing numbers with the original bank code before migration.

Conclusion:

In this experiment, I learnt that the Bank Management System database was successfully designed using the Entity-Relationship (ER) model. The database identifies the major entities, their attributes, primary keys, and relationships required for managing banking operations. The ER diagram provides a clear conceptual representation of the system and serves as the foundation for designing an efficient and reliable relational database. The experiment enhanced the understanding of ER modeling and its application in real-world database design.

Signature of faculty in-charge